# Práctica dos: Sistema Cardiovascular

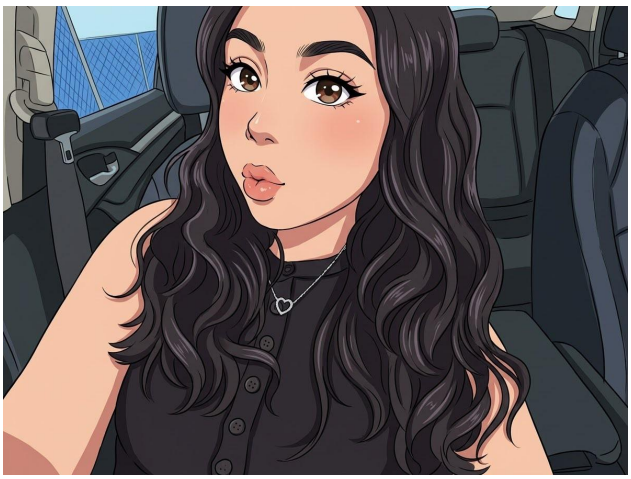
**Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica**

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## Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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## Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('sistemab1')
```

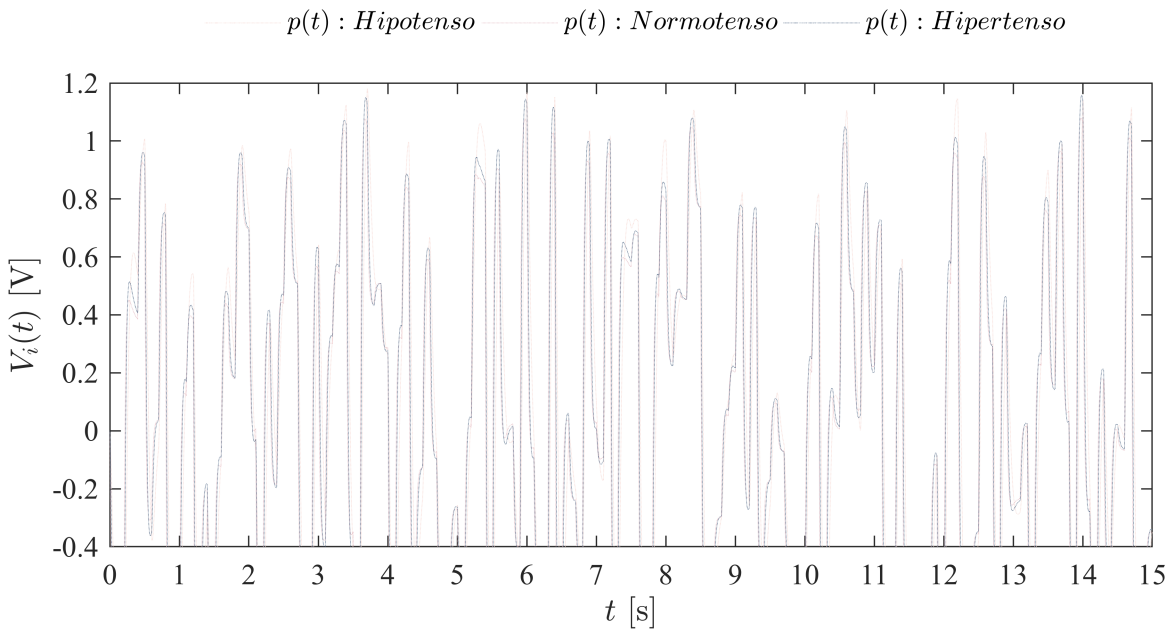
```
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-2';
x= sim('sistemab1',parameters);
```

```
Found algebraic loop containing:
sistemab1/Sistema: Normotenso/Pp(t): Modelo de EID/RZ//(Z+R)
sistemab1/Sistema: Normotenso/Pp(t): Modelo de EID/C
sistemab1/Sistema: Normotenso/Pp(t): Modelo de EID/Derivative
sistemab1/Sistema: Normotenso/Pp(t): Modelo de EID/Add (algebraic variable)
```

```
t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data,x.Pa,t, 'previous');
writematrix(x.Pa,'signal.xlsx');
```

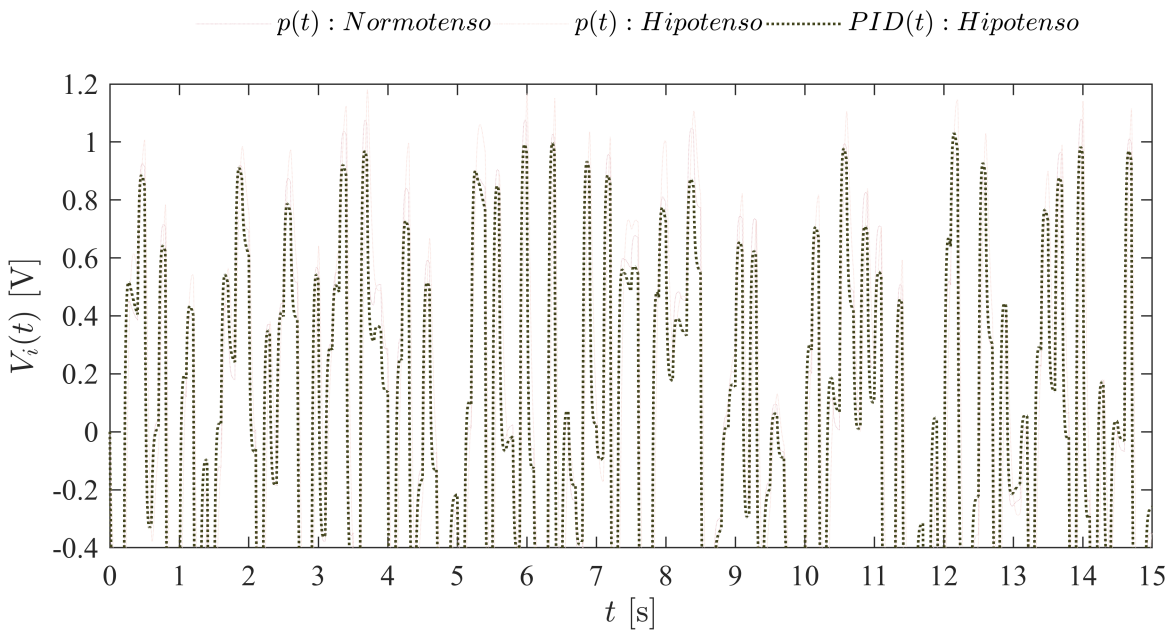
## Respuestas: Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t,x.PpL,x.PpN,x.PpH,'individuos')
```



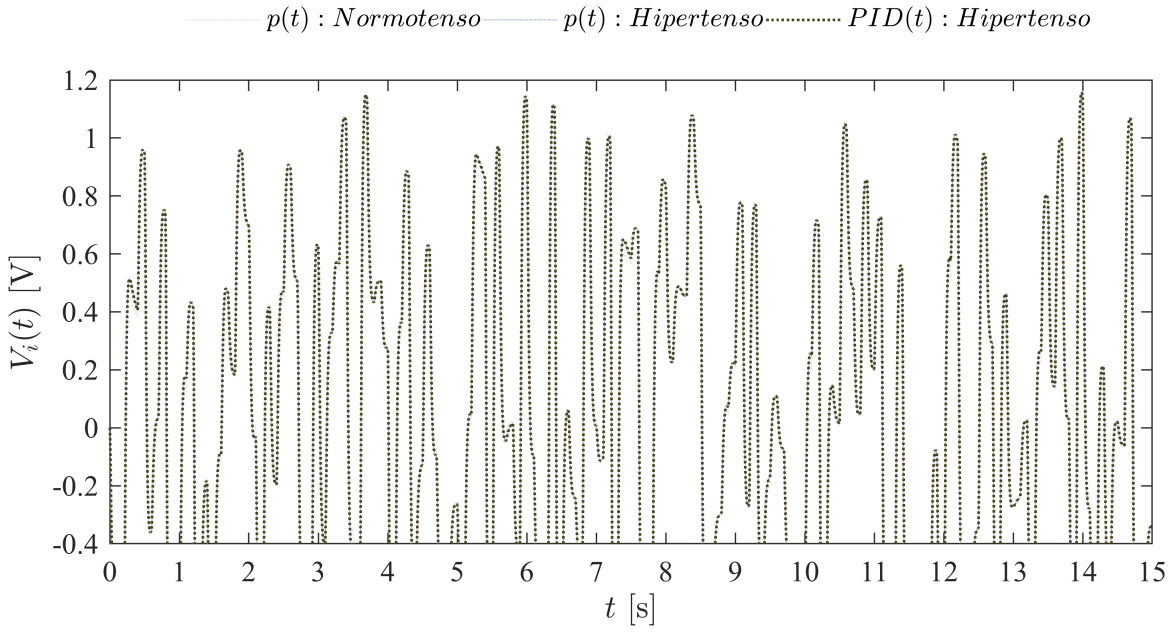

## Respuesta : Normotenso,  Hipotenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpL,x.PIDL,'hipotenso')
```




## Respuesta : Normotenso,  Hipertenso y Controlador

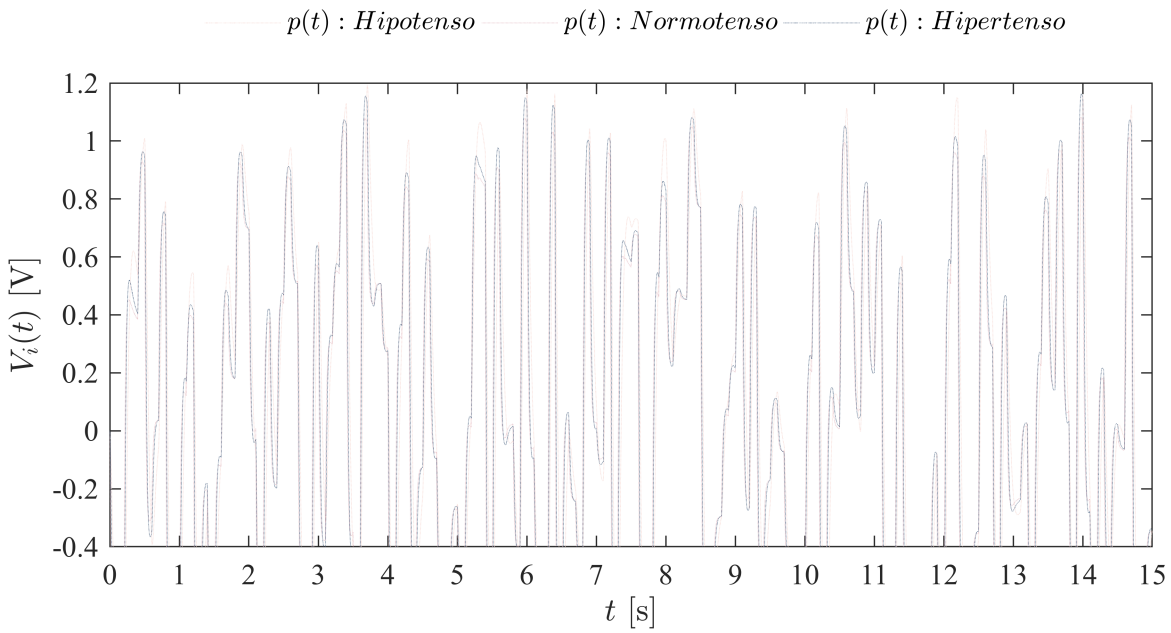
```
plotsignals(x.t,x.PpN,x.PpH,x.PIDH,'hipertenso')
```




## Modelo de EDOs

```
par_hipo = [0.01,0.001,1.00,3.500];
par_normo = [0.01,0.001,1.00,1.500];
par_hiper = [0.01,0.001,1.00,1.000];
```

```
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpL,PpN,PpH,'odes');
```




```
function Pp = cardio(par,Pa,t)

  Z = par(1); L = par(2); R = par(3); C = par(4); dt = t(2);

  n = round(t(end)/dt);

  FL = zeros(length(t),1); FL(1) = 0;
  Pp = zeros(length(t),1); Pp(1) = 0;

  for i = 1:n
      FL(i+1) = FL(i) + (-Pp(i)/L + Pa(i)/L)*dt;
      Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/(Z*C))*dt;

  end

end
```

## Funcion: Respuesta a la simulacion

```
function plotsignals(t,x1,x2,x3,caso)
    set(figure(),'color','w')
    set(gcf,'units','inches','position',[1,1,7,3.5])
```

```
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = [247, 214, 208]/255;
    clr2 = [226, 180, 189]/255;
    clr3 = [62, 88, 121]/255;
    clr4 = [65, 67, 27]/255;


  if caso == "individuos"
    plot(t,x1,':','LineWidth',0.25,'color',clr1) % Hipotenso
    plot(t,x2,':','LineWidth',0.25,'color',clr2) % Normotenso
    plot(t,x3,':','LineWidth',0.25,'color',clr3) % Hipertenso
    L = legend('$p(t):Hipotenso$','$p(t):Normotenso$','$p(t):Hipertenso$');

elseif caso == "hipotenso"
    plot(t,x1,':','LineWidth',0.25,'color',clr2) % Normotenso
    plot(t,x2,':','LineWidth',0.25,'color',clr1) % Hipotenso
    plot(t,x3,':','LineWidth',1,'color',clr4) % Controlador del hipotenso
    L = legend('$p(t):Normotenso$','$p(t):Hipotenso$','$PID(t):Hipotenso$');

elseif caso == "hipertenso"
    plot(t,x1,':','LineWidth',0.25,'color',clr2) % Normotenso
    plot(t,x2,':','LineWidth',0.25,'color',clr3) % Hipertenso
    plot(t,x3,':','LineWidth',1,'color',clr4) % Controlador del hipertenso
    L = legend('$p(t):Normotenso$','$p(t):Hipertenso$','$PID(t):Hipertenso$');

elseif caso == "odes"
    plot(t,x1,':','LineWidth',0.25,'color',clr1) % Hipotenso
    plot(t,x2,':','LineWidth',0.25,'color',clr2) % Normotenso
    plot(t,x3,':','LineWidth',0.25,'color',clr3) % Hipertenso
    L = legend('$p(t):Hipotenso$','$p(t):Normotenso$','$p(t):Hipertenso$');
    end


set(L,'interpreter','latex','location','northoutside','orientation','horizontal',
'box','off')
    xlabel('$t$ [s]','interpreter','latex')
    ylabel('$V_i(t)$ [V]','interpreter','latex')

    xlim([0 15]); xticks(0:1:15)
    ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

end
```